



SI100B: Introduction to Information Science and Technology

Lecture 0: Introduction

School of Information Science and Technology (SIST), ShanghaiTech University

Fall Semester, 2025





Course Objectives & Design



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- Learn some basic knowledge of IST
- Three parts:
 - Python Programming & AI (9 weeks)
 - Basic principles of computer programming
 - Introduction of AI
 - Electronic Engineering (3 weeks)
 - Fundamental semiconductor components of the information age
 - How electronic technology achieves computing and communication with the physical world;
 - Typical systems based on electronic technology.
 - Course Projects (4 weeks)
 - ~6 themes focusing on PP or EE aspects.
 - Choose one of the topics in groups and conduct a 4-week course project



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Grading policy



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- CS part: 50%
 - 3 Programming homework 15%
 - Computer-based Exam: 35% @Week 9
- EE part: 20%
- Course project: 25%
- Class Participation & Quiz: 5%



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Instructors

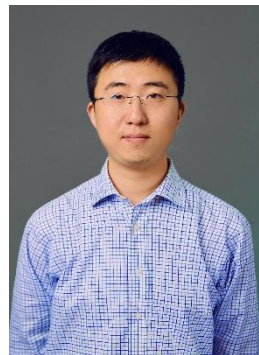


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何旭明

Python Programming & AI



屠可伟

Electronic Engineering



娄鑫

陆林燕

刘闯

李正浩

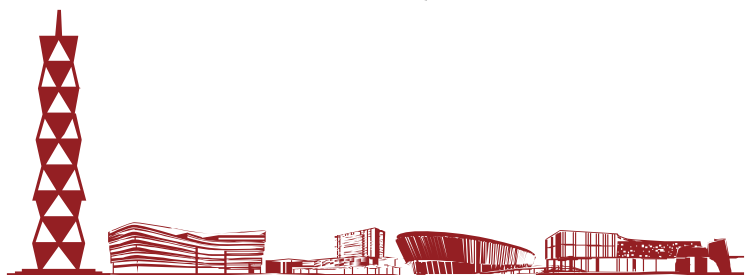
白卫邦

林竹雨

娄鑫

Course Project

- Instructor office hours:
 - Booking with instructors via email
 - {hexm, tukw, louxin}@shanghaitech.edu.cn



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- 16 Teaching Assistants
- Tutorial Sessions:
 - Friday: class 11-12, 18:55-20:35, Week 1-9
 - Location: this classroom
- Office Hours:
 - To be announced in BB & Piazza
 - Please participate in the time slot survey





Course Systems



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- Main course page (on Blackboard 互动教学平台)
 - Lecture slides
 - Homework assignments
 - Project description
 - Reference/additional reading materials
 - <https://elearning.shanghaitech.edu.cn:8443>
- Piazza
 - Online discussion and QA
 - <https://piazza.com/shanghaitech.edu.cn/fall2025/si100b>
- OJ
 - Programming homework submission and grading



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Communication



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- For technical discussions:
 - Face-to-face meetings during tutorials or office hours
 - Emails should be used only for short communication (e.g. making an appointment), but not for technical discussions
 - Use Piazza forum



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Academic Integrity



- Ethical policy of academia
 - avoidance of **cheating or plagiarism**;
 - maintenance of academic standards;
 - honesty and rigor in research and academic publishing;
- Dos
 - all work should be done **individually and independently**;
- Don'ts
 - **copy** from the web or others (including **AI tools**);
 - **give or expose** your code to others;
 - make your code **public**;
- We
 - have powerful tools to **automatically detect** code plagiarism;
 - zero tolerance of any misconduct;

*Use homework to
practice for midterm
exam!*

Details: Read 《上海科技大学学术道德规范（试行）》 《信息学院学术诚信问题的处罚细则（试行）》



- Plagiarism punishment
 - When one student copies from another student, **both** students are responsible
 - Up to 100% score penalty on the assignment or exam in question
 - Repeated violation will result in an F grade for this course as well as further discipline at the school/university level





Transiting from high school to college

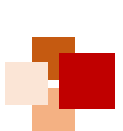


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- Be **self-driven**, not teacher-driven or parent-driven



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AI coding tools

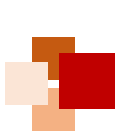


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- Why are you still learning programming when LLMs can write code "better" than you?
- Reason 1: learning arithmetic vs. using calculator
- Reason 2...



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A real experiment on AI tool use

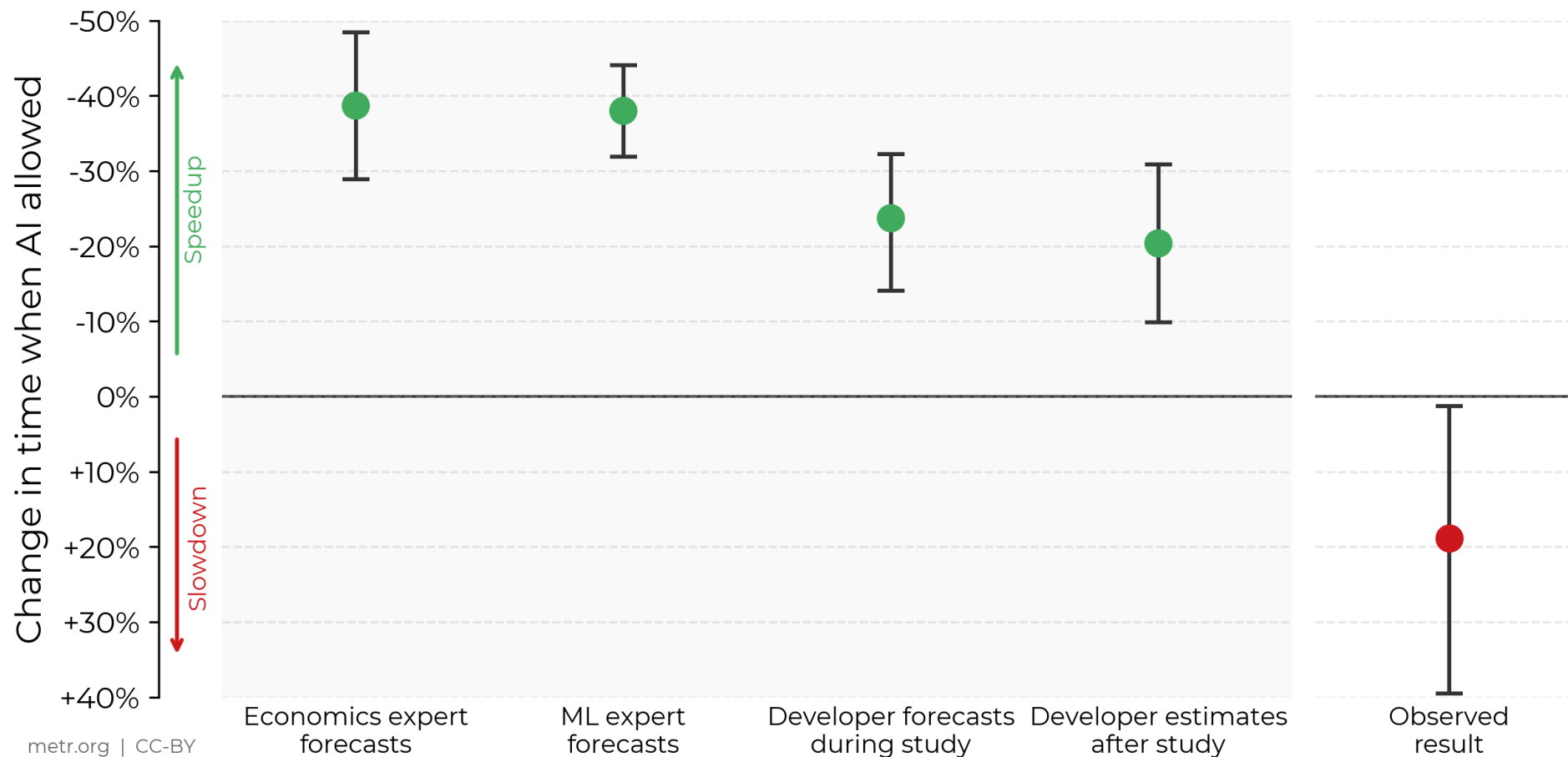


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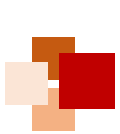
Against Expert Forecasts and Developer Self-Reports, Early-2025 AI Slows Down Experienced Open-Source Developers



In this RCT, 16 developers with moderate AI experience complete 246 tasks in large and complex projects on which they have an average of 5 years of prior experience.



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Computer Proficiency Survey



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选项	小计	比例
I do not have much experience of using computers.	47	19.34%
I have experience of using computers but have never learned to write programs.	54	22.22%
I have learned very basic programming but I am not good at it.	116	47.74%
I am good at programming but perhaps not at the competition level.	20	8.23%
I have won awards in programming competitions.	6	2.47%

A班

B班

可申请免听



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